

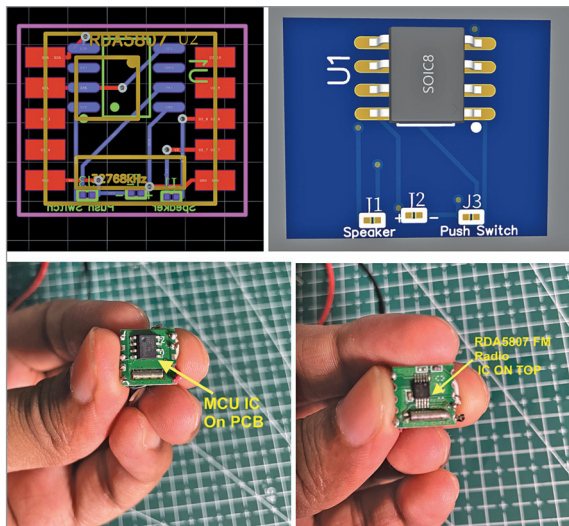


Fig. 9: Top layer showing FM module PCB design; bottom layer showing MCU IC and jumper connector mounting

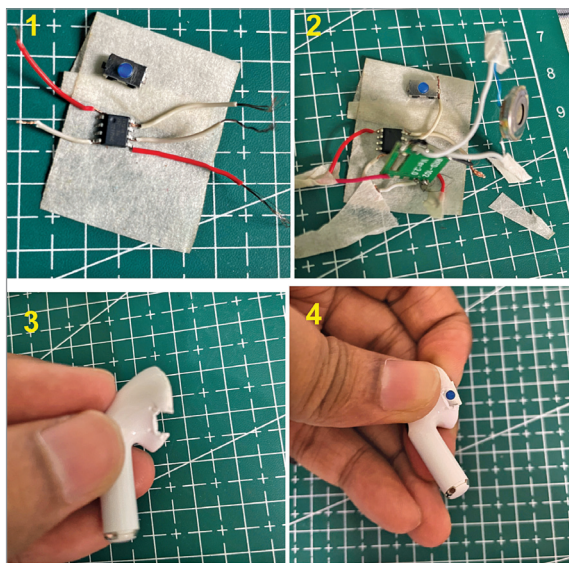


Fig. 10: SMD soldering and connection

circuit diagram of the system.

The circuit is powered by a 3.3V battery and controlled by a slide switch (SW1). This SPST switch connects the battery to the MCU (pin 1) and FM module (pin 5) when switched on. Sliding it to the off position disconnects power.

The MCU communicates with the FM module via I²C: PA1 for SDA and PA2 for SCL. A pushbutton switch (SW2) connected to PA3 manages channel and volume

control according to the programmed logic.

Audio output from the RDA5807's L-OUT and R-OUT pins is routed through a 2-pin SMD pad (J1) to a speaker, with proper grounding maintained throughout.

After designing the circuit, testing may be performed by:

Soldering components directly using SMD soldering (as seen in Figs. 5 and 7), or

Designing a custom PCB

In this system, SMD soldering was used. Wires were soldered directly to the MCU and FM module pins, and then interconnected according to the circuit diagram. A cut-out was made in the earbud casing to accommodate the switch.

Note. A pre-built earbud case was used. However, a custom-designed enclosure may be created to accommodate a higher-capacity battery, depending on the

intended application.

PCB design

Fig. 9 shows the top layer showing FM module PCB design and the bottom layer shows MCU IC and jumper connector mounting. A better, neater, and more compact approach is to design a PCB of the same tiny size, with the FM radio module surface-mounted on the top layer and the MCU on the bottom layer. Such placement is followed

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in this PCB design. Three pairs of solder pads are also provided for the TWS speaker connection, battery connection, and switch connection.

Fig. 9 shows the PCB design, which is approximately the same size as the FM radio module. The next image displays the bottom layer with the MCU and SMD connector pads, followed by an image illustrating how the MCU fits neatly on the back side of the FM module on the PCB. Fig. 10 shows SMD soldering and connections.

Testing

To begin testing the FM AirPod system, slide the SPST switch (SW1) to the ON position. This powers both the ATtiny412 microcontroller and the RDA5807 FM module. Once powered, use the pushbutton (SW2) connected to PA3 to perform various functions based on the duration of the press. A quick press (less than 2 seconds) seeks the next available radio channel. Holding the button for 2 seconds seeks the previous channel. Holding it for 10 seconds increases the volume, with a maximum limit of 15. Holding the button for 30 seconds decreases the volume, down to a minimum of 0. All changes made through the pushbutton interface are automatically saved to the EEPROM memory. **EFY**

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